INCH-POUND
MIL-DTL-24643/48E
22 August 2002
SUPERSEDING
MIL-C-24643/48D
22 November 1994

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, 1000 VOLTS, TWO CONDUCTOR, TYPE LSDNW

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24643 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First -	Copper conductor, uncoated (see table I for size).
Second -	Separator (opaque white) may be used at manufacturer's option where required to provide free-stripping insulation.
Third -	Cross-linked polyethylene insulation or ethylene propylene rubber insulation (see table I for thickness). Standard identification code shall be applied by method 1 or method 3. If identification method 1 is used, insulation color shall be black.
Fourth -	Two conductors cabled together with a lay not greater than 24 times the pitch diameter. Fillers shall be used as necessary to form a firm, well rounded assembly.
Fifth -	Binder tape applied helically with overlap.
Sixth -	Cross-linked polyolefin jacket (see table I for thickness).

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.
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TABLE I. Details.

Military part no. M24643/48	Size	AWG conductor size	Insulation Thickness min avg (inch)	Cable jacket thickness min avg (inch)	Overall diameter (inches) TYPE LSDNW	
					min	max
-01UN	3	16 (class B)	0.030	0.040	0.361	0.390
-02UN	4	14 (class B)	0.030	0.040	0.398	0.430
-03UN	9	10 (class B)	0.030	0.040	0.504	0.545
-04UN	14	9 (class B)	0.045	0.040	0.564	0.610
-05UN	23	7 (class B)	0.045	0.045	0.638	0.690
-06UN	50	3 (class C)	0.045	0.050	0.842	0.910
-07UN	75	1 (class C)	0.055	0.055	1.000	1.080
-08UN	100	0 (class D)	0.055	0.055	1.082	1.170

TABLE II. Electrical details.

Size	AWG conductor size	Conductor resistance (ohms)	Insulation resistance (megohms)	Voltage withstand (minimum) (volts)	Cold bending mandrel diameter (max inch)	Accelerated service loading current (amperes)
3	16 (class B)	4.3	6000	4000	4	----
4	14 (class B)	2.68	5000	4000	4	----
9	10 (class B)	1.08	3800	4000	5	----
14	9 (class B)	0.86	4400	5500	6	100
23	7 (class B)	0.54	3700	5500	7	130
50	3 (class C)	0.21	2600	5500	9	215
75	1 (class C)	0.13	2600	7000	10	285
100	0 (class D)	0.11	2400	7000	11	335

EXAMINATION AND TESTS:

Basic electrical: Requirements

Conductor resistance - ohms/1000 feet at 25°C, max ..... (see table II)
Voltage Withstand - volts, root mean square, minimum ..... (see table II)
Insulation resistance - Conductor to conductor
Megohms -1000 feet minimum ..... (see table II)
Conductor continuity ..... No failure

Group A:
Visual and dimensional..... No failure

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Group B:
- Cross-linked proof test (percent, maximum)
  - Insulation 50
- Jacket (when tested at 200°C) 50
- Cold bending, cable - (see table I for mandrel diameter) ..... No damage
- Tear - pounds per inch thickness, minimum (ASTM D 470) 35

Physicals (unaged)
- Insulation
  - Ethylene propylene rubber
    - Tensile strength - lb/in^2, minimum 700
    - Elongation - percent, minimum 250
  - Cross-linked polyethylene
    - Tensile strength - lb/in^2, minimum 1800
    - Elongation - percent, minimum 250
- Jacket (cable)
  - Tensile strength - lb/in^2, minimum 1300
  - Elongation - percent, minimum 160

Group C:

Physicals (aged)
- Insulation
  - Ethylene propylene
    - Air Oven
      - Tensile strength - percent of unaged, minimum 75
      - Elongation - percent of unaged minimum 75
  - Cross-linked polyethylene
    - Air Oven
      - Tensile strength - percent of unaged, minimum 80
      - Elongation - percent of unaged minimum 80
- Jacket (cable)
  - Air oven
    - Tensile strength - percent of unaged, minimum 60
    - Elongation - percent of unaged minimum 60
  - Hot oil immersion
    - Tensile strength - percent of unaged, minimum 50
    - Elongation - percent of unaged minimum 50
- Shrinkage ..... No failure
- Heat distortion - percent of unaged, maximum 30
- Permanence of printing (conductor) - method 1 only, cycles, minimum 25
- Permanence of printing (jacket) - cycles, minimum 125

Group D:
- Flame propagation (cable) ..... No failure

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QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements
Aging and compatibility (cable)(125 ± 5°C)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS:

Type and Size	Feet (nominal)
LSDNW-3	2500
LSDNW-4 through 23	2000
LSDNW-50	1500
LSDNW-75, 100	1000

Note: Not for AIR FORCE use.

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Custodians:
Army - MI
Navy - SH

Review Activities:
Army - AV, CR
Navy - CG, EC
DLA – CC

Preparing Activity:
Navy - SH
(Project 6145-2308-046)

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